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Research paper

JobEdKG: An uncertain knowledge graph-based approach for recommending online courses and predicting in-demand skills based on career choices

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ABSTRACT

Modern job markets often require an intricate combination of multi-disciplinary skills or specialist and technical knowledge, even for entry-level positions. Such requirements pose increased pressure on students, new graduates, and job seekers to find suitable sources of information to enhance their employability. Unlike standard competence management systems, this paper presents JobEdKG, which helps job seekers from various backgrounds choose online courses based on their prospective careers in addition to predicting different skills that may be required in their chosen career path. While existing solutions focus on internal institutional data, such as previous student experiences and a fixed set of skills provided by curated datasets, JobEdKG considers external data, recommending online courses that best cover the knowledge and skills required by selected job roles, in addition to extracting and predicting skills of that particular job roles. To achieve this, we first extract skills from job postings and online courses. These skills are linked to job titles, online courses, and other concepts in order to create a knowledge graph (KG). We assign an uncertainty score to each fact in the KG based on the prevalence of the fact in the source data (i.e. job listings and online courses), which results in an uncertain KG (UKG). Finally, we model the constructed UKG in order to infer different relations between the different concepts. The code and the data are available on our GitHub repository (https://github.com/team611/JobEd) and a user interface to browse the KG is available at (http://jobed.datanets.org/).

1. Introduction

In recent times, several significant trends have emerged in the job market. First and foremost, the number of job changers has been on the rise (Borowczyk-Martins and Pacini, 2022), and the average duration of employment in a single position has shortened (Stat, 2019). Secondly, globalization has widened the pool of potential job candidates for each position, in addition to candidates, on average, possess higher levels of education compared to previous years (Akçomak et al., 2011). These factors collectively contribute to a highly competitive labor market (Antràs et al., 2006).

Moreover, the evolving demands of the job market, coupled with the rapid digitization of various industries (Peng, 2017), have made computer skills increasingly valuable. The COVID-19 pandemic has further compounded this by accelerating technological adoption while leading to workforce reductions in some sectors (World Economic Forum, 2020). Additionally, soft skills, such as interpersonal abilities, are also gaining prominence (Borghans et al., 2014).

In order to keep up with this fast-paced evolution, higher education institutions have made use of technology to increase the adaptability of their teaching and learning techniques (Santandreu Calonge and Aman Shah, 2016). Massive Open Online Courses (MOOCs) have become highly popular in the domain of online education. These programs are primarily supplied by elite colleges and universities, with 88% of all MOOCs offered by institutions that are ranked in the top 50 by US News & World Report (Santandreu Calonge and Aman Shah, 2016). There seems to be a noticeable increase in both the number of MOOC users and the number of MOOCs being given by various platforms and courses (Gamage et al., 2015). Companies are increasingly curious to see whether MOOCs may assist close the skills gap among their newly hired university graduates as the popularity of these courses continues to soar (Santandreu Calonge and Aman Shah, 2016).

1.1. Problem statement and research scope

Knowing which talents are pertinent, in demand, and supply, as well as emphasizing the courses that would help job searchers acquire these

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abilities, is crucial for facilitating the matching of candidates to jobs. Competency management systems (CMSs) (Judrups et al., 2015) are some of the earliest systems that were developed to solve this problem. They are used by companies to profile the different skills that their employees have and the skills that different training programs offer. However, these systems suffer from multiple limitations that hinder their applicability in multiple cases. First, they employ databases with rigid schemas and internal proprietary data resulting in their limited accessibility to a wider audience. They base their decisions on employees, not job profiles, making them non-transferable. They are embedded within the structure of the company, making them inapplicable to varying contexts. Updating these databases can be labor-intensive and costly since they need to be manually curated. The effectiveness of these systems lies in the diversity and breadth of skills that they represent, needing to be updated consistently by manual annotators, which is particularly difficult because of the dynamic nature of the job market. All of these limitations and the current advancements in machine learning inspired the creation of multiple data-driven approaches that are less expensive to employ and take advantage of existing educational resources and public data.

Knowledge graphs (KGs) are one of the newest and most promising data structures in the field. KGs offer a structured representation of the relationships between competencies, roles, and learning resources, enabling a holistic view of the labor market's skill ecosystem. Machine learning techniques harness this data to analyze patterns, predict skill gaps, and recommend tailored learning pathways. However, most existing approaches that employ KGs use them as exploratory search tools to gather insights using precise queries (Ashok et al., 2018; Galkin et al., 2018), which does not take into consideration KGs inherent incompleteness. KG completion can be useful in the discovery of new relationships between entities. For example, new jobs have limited information available, in order to mitigate this, KG completion can be used to uncover the new jobs' relation to existing skills, training programs, and courses.

1.2. Research objectives

In this paper, we aim to create and model a probabilistic knowledge graph for labor market entities and online courses (JobEdKG) based on data derived from online job postings and available online courses. We aim to provide employers, policymakers, and job seekers with a public resource to explore the skill demands in the job market and help remedy the skill gap by linking these skills to courses that will help acquire them. Additionally, we perform link confidence prediction using uncertain KG embedding models to predict the probability score of each link on the constructed KG. Our pipeline for link confidence prediction enables the inference of new connections in the knowledge graph (KG) and serves as a valuable resource for potential applications such as predicting skill demands and recommending courses. This paper makes multiple significant contributions.

- To the best of our knowledge, we are the first to construct a large-scale KG that combines labor market and education entities. Additionally, we also present a construction pipeline for our KG, which can provide guidelines for KG construction of other labor markets in multiple countries.
- We propose a novel fully unsupervised approach for skill entity recognition from job offers. This approach has the potential to be utilized in various other digital domains and effectively addresses the issue of insufficient annotated data.
- We perform link confidence prediction on JobEdKG and we leverage the results for labor market skill demands prediction and course recommendation.

1.3. Paper organization

The organization of this paper is outlined as follows: Section 2 provides an examination of the related literature. Section 3 outlines our KG construction methodology, which includes the construction process, the quality evaluation and the description and visualization of the resulting KG. We describe and implement details of our KG completion model in Section 4. The experiments we conducted are presented in Section 5 where we also showcase the results of our case study. We discuss further implications and future work in Section 6. Finally, we conclude the paper in Section 7.

2. Related work

In order to fully understand KGs and KGs in the field of job market analysis and learning, it is important to examine the existing research on this topic. Therefore, this section will provide a comprehensive review of the related work on knowledge graphs in education and employability and skill extraction.

Competency management systems (CMSs) are the earliest incarnation of this line of research. They match employee profiles to different courses and training programs that offer skills that are lacking in the company. These systems are known for their rigidity, since they are constructed using manually curated databases of proprietary data, making them unfit to employ in an ever-changing job market. This motivated the creation of different data-driven approaches that can take advantage of publicly available knowledge in order to personalize learning to different job seekers. Knowledge graphs (KGs) are one of the leading data structures in this regard, attributed to their ability to structure complex information facilitating easy exploration of existing knowledge and discovery of new knowledge.

2.1. Knowledge graphs in job market analysis and personalized learning

Methodologies for employing KGs to improve education and its results have been proposed in a number of research papers. In one study, the authors of (Chen et al., 2018) established a KnowEDU system that creates educational KGs automatically. The resulting KG is made up of nodes that stand in for educational topics that students should be able to master. The CRF model, BiLSTM, and the Bidirectional Encoder Representations from Transformers (BERT) were combined in a different study to create a method for the autonomous generation of an educational KG (Su and Zhang, 2020). They concentrated on developing an educational KG that gathers essential data and captures relevant information. Another educational KG was constructed in (Li et al., 2022).

In the realm of personalized learning, the system in (Huang, 2011) was built using knowledge structure graphs. To give students individualized learning materials based on the created KG, the authors created and implemented an optimized learning route generator. Another useful and relatively recent strategy for enhancing education is to use KGs in interactive applications.

Studies such as (Luo et al., 2019) have tackled the issue of employment representation in KGs. In order to capture the similarity and ordering relationships between job titles and business pairings, the authors created a representation of employment roles. The Skills & Occupation KG developed in (de Groot et al., 2021) is the most recent study in this field of job advice. This KG was created by incorporating external job ads into the taxonomies of skills and occupations that already existed.

Another study created a KG to represent the artificial intelligence job market and the accompanying necessary abilities (Jia et al., 2018). The KG made it easier to investigate the connections between various AI-related key competencies.

2.2. Knowledge graph embeddings

In the studies mentioned in the previous section, KGs act as structured representations of entities and their relationships in the real world. KGs can be divided into two categories: Deterministic KGs, which consist of relation facts that depict semantic connections between entities, and Uncertain KGs, which assign confidence scores to relation facts to indicate the degree of certainty regarding their accuracy.

Embedding models are essential for integrating structured knowledge representations into machine learning. Recently, several methods have been introduced to embed Uncertain Knowledge Graphs (UKGs) and facilitate automated knowledge acquisition. One such method, called UKGE (Chen et al., 2019), models the plausibility of triples by multiplying embedding vectors and mapping the result to a confidence score ranging from 0 to 1.

Another method, UOKGE (Boutouhami et al., 2020), improves confidence prediction by jointly encoding graph and ontology structures. However, UOKGE requires an additional ontology of entity types, which may not be available for all KGs. Another approach, URGE (Hu et al., 2017), uses matrix factorization to embed uncertain graphs but only considers node proximity in the networks. Recent advancements in embedding methods strive to represent elements using more intricate geometric structures rather than simple vectors such as (Vendrov et al., 2015) that uses infinite cones to represent graph nodes and (Vilnis et al., 2018) that represents elements using axis-aligned hyperrectangles, or boxes. Several methods have been proposed to improve the training of box embeddings, including smoothing techniques (Li et al., 2018) and methods to enhance their performance (Dasgupta et al., 2020; Chen et al., 2021).

In summary, these studies have made some progress in constructing educational and labor market KGs but they did not give a general framework to construct a KG that combines both educational and labor market entities in order to help analyze both components. Furthermore, no studies employed uncertain KG embeddings for link confidence prediction for education and labor market KGs. Such contribution is critical since the level of importance of entities in regard to each other is crucial for better data representation.

2.3. Skill extraction

Skill extraction from text has been tackled significantly throughout the literature (Khaouja et al., 2021). Studies such as (Khaouja et al., 2019) developed a methodology for soft skill extraction based on word embeddings and social network analysis. (Li et al., 2020) utilized their large taxonomy with 30k job titles and 50k skills for skill entity tagging then used FastText (Bojanowski et al., 2017) to capture the semantic understanding of job-related information, while employing entity embeddings to represent the consistency of skills within the job.

The skill extraction module in (Gugnani and Misra, 2020) utilized various natural language processing (NLP) methods to detect skill-related terms starting with named entity recognition, part of speech tagging and models like word2vec (Mikolov et al., 2013) and evaluating their scores using skill dictionaries. Other studies such as (Tamburri et al., 2020) utilized data annotation and Transformer-based models such as BERT for skill extraction. In order to address the scarcity of available tools for extracting skills from job descriptions, multiple studies have chosen the direction of developing annotated datasets for skill extraction. (Beauchemin et al., 2022) introduces a novel public dataset called FIJO, which comprises insurance job offers and includes annotations for numerous soft skills. The study proceeds by presenting the outcomes of skill detection algorithms utilizing a named entity recognition approach, demonstrating that models based on transformers exhibit excellent token-wise performance on this dataset. Other papers introduced the dataset SKILLSPAN (Zhang et al., 2022b), which includes 14.5 K sentences and more than 12.5 K annotated spans. The study also provided guidelines for this dataset, which were developed by domain experts and based on annotations from three distinct sources. These guidelines cover both hard and soft skills. (Green et al., 2022) also introduced a dataset that includes 18.6k entities categorized into five types: Skill, Qualification, Experience, Occupation, and Domain. In the same realm, (Zhang et al., 2022a) released the first Danish skill classification dataset with distantly supervised fine-grained labels using the ESCO taxonomy. The authors used zero-shot and few-shot classifications to evaluate their dataset. (Ternikov, 2022) carried out a study where the authors identified skills by employing a combination of skill standardization, hierarchical clustering, and association mining techniques.

Though these studies have several advantages, they rely heavily on data annotation and supervised learning which is an issue if we are dealing with languages and sectors that do not have many annotated data sources. Furthermore, these studies targeted skill extraction from job ads and not necessarily from courses, which is a vital step to link the labor market and education.

3. Knowledge graph construction

We propose an automated pipeline that encompasses four key modules: data collection and processing, construction of our knowledge

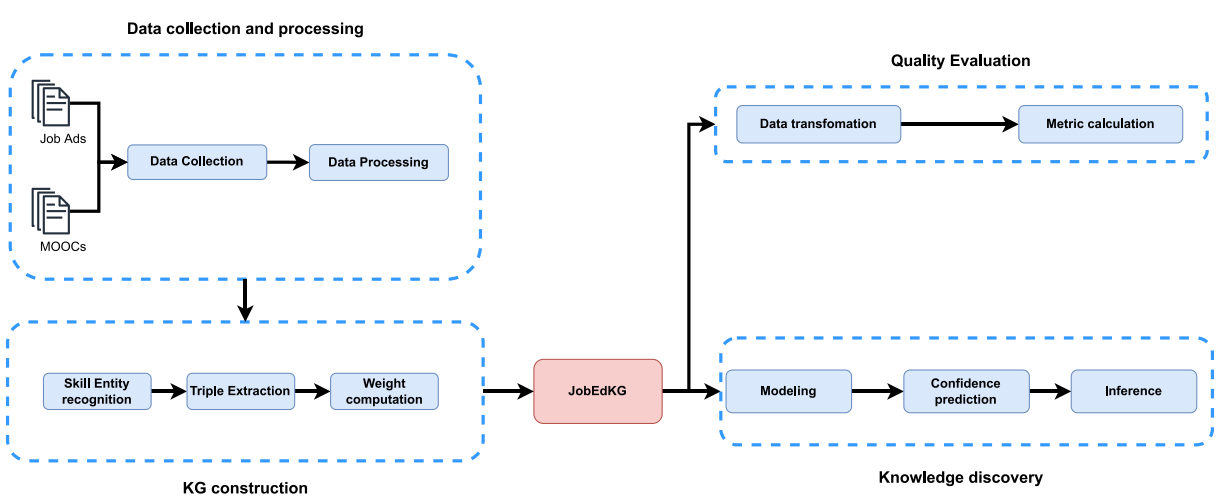


Fig. 1. Overview of our methodology.

graph, evaluation of KG quality, and knowledge discovery. In this subsection, we describe the methodology we employed for constructing our KG, which involves several sequential steps. Fig. 1 showcases these steps.

3.1. Data collection

In the course of this study, we collected job postings from the Moroccan website Rekrute.com[1] using Python. Each job posting contains the job title, cognitive skills (hard skills), non-cognitive skills (soft skills), the recruiter company, the economic sector to which the job title belongs, the city where the job is located, the required degree, the required experience, the type of contract, the function of the job, the date the job offer was posted and the date the job offer expires. We collected job postings from November 2005 up until September 2022. To avoid the duplication of job postings, we extract the recruiter companies' identifiers from the website in order to verify if a job posting already exists in our dataset. We only keep the job postings that offer a job in Moroccan cities. The final dataset contains a total of *99676 job postings* in both English and French.

We also collect MOOCs from the website Coursera[2] using the website's API. The final MOOCs dataset contains a total of *10180 courses* with the names of the courses, descriptions, specializations, and instructors.

3.2. Skill entity recognition

In order to extract hard skills from the different job postings and course descriptions we opted for a rule-based approach that uses existing databases of hard skills and extracts skills using the Spacy library rule mining module.[3] Procedure 1 describes the rule building and the entity extraction methodologies. We extracted hard skills in both French and English. For the hard skills mentioned in English, we used an existing set of rules from Jobzilla.[4] For the hard skills mentioned in French, we constructed our own set of rules based on the ESCO[5] and ROME[6] hard skills datasets. Step 1b of Procedure 1 describes the rule-building process from raw skills data where, for each hard skill, we applied an exact word-building process and a POS (Part-of-speech) tagging-based process. Fig. 3 showcases an example rule building from a hard skill using both exact word-based rule building and POS-tagging-based rule building.

We opt for rule extraction instead of model-based extraction for multiple reasons seeing that the entities extracted using rules are inherently normalized since they are based on a preexisting curated dataset. Additionally, model-based extraction needs a significant amount of annotated data to get reliable results and is less computationally efficient.

After extracting the hard skills using the rules, we identify the incompleteness of a significant number of them, so we apply sentence filtering using the TextAttack model.[7] Seeing that this model is trained on English data, we translated the extracted skills using two translators: Argos translator.[8] and a transformer-based translator[9] Argos translator takes more liberty in the translation depending on the semantic and syntactic correctness of the input sentences, meaning that it deviates from the original sentence's literal translation depending on how structurally sound and meaningful the sentence is. The transformer-based translator is a direct translator that produces literal translations of sentences to have a correct syntax while preserving the original semantics even if they are not sound.

After translating the skills, we use the pre-trained TextAttack model to identify if the sentence (skill in our case) is complete or not in addition to quantifying the degree of their completeness with an acceptability score. The skill is kept if all three following conditions exist: both translations are labeled as a complete sentence, the difference between the acceptability scores of both translations is less than 0.15 (to ensure that the sentences are syntactically and semantically sound) and the acceptability score is above 60% for at least one of the translations. The methodology for skill entity recognition is illustrated in Fig. 2.

Procedure 1: Rule building and entity extraction process for hard skill named entity recognition (NER)

1. Rule building
   - (a) Source data collection:
     - i. Rule data collection (Jobzilla)
     - ii. Raw skills data collection (ESCO)
   - (b) Rule building from raw skills data: for each hard skill
     - i. Exact word rule building
       - A. Segment hard skill to multiple segments in increasing order
       - B. For each segment: Exact word rule building
     - ii. POS-tagging-based rule building
       - A. Extract POS tags of skill
       - B. Create POS-taging-based rule: leave the first original word exactly and replace other words with their POS.
2. Entity extraction
   - (a) Using the constructed rules extract hard skills.
     - i. Set the rule database.
     - ii. Extraction process:
       - A. Exact word rules extract a hard skill-based of word similarity
       - B. POS rules extract the first word based on word similarity, and the other words are extracted based on POS tag similarity.
   - (b) Check the semantic correctness of hard skills using the linguistic acceptability model.
     - i. The skill is translated into English using two translators.
     - ii. Every translation is classified using a linguistic acceptability classifier.
     - iii. A hard skill is accepted if both translations are linguistically acceptable, and the probability of acceptance is above 70% for both of them.

3.3. Triple extraction and weight assignment

Relations are extracted based on the co-occurrence of different types of entities in the same job offer or the same course. For example, if a job offer describes a job title belonging to a certain sector, then that job title and sector are linked. The outcomes of skill entity recognition

[1] https://www.rekrute.com
[2] https://www.coursera.org
[3] https://spacy.io/usage/rule-based-matching
[4] https://github.com/pandmi/jobzilla_ai
[5] https://esco.ec.europa.eu
[6] https://www.pole-emploi.fr/employeur/vos-recrutements/le-rome-et-les-fiches-metiers.html
[7] https://huggingface.co/textattack/roberta-base-CoLA
[8] https://github.com/argosopentech/argos-translate
[9] https://huggingface.co/Helsinki-NLP/opus-mt-fr-en

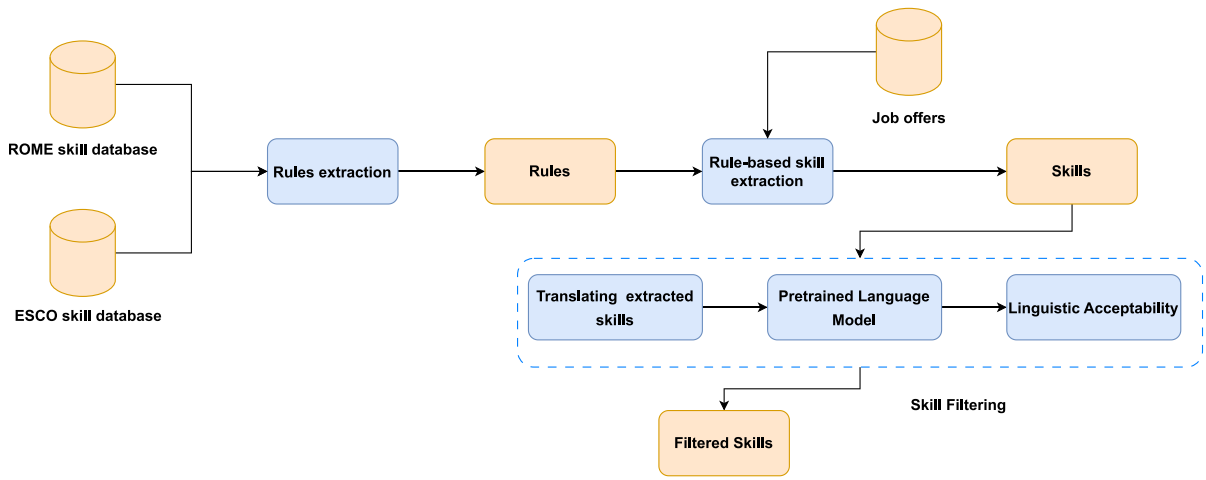


Fig. 2. Skill entity recognition.

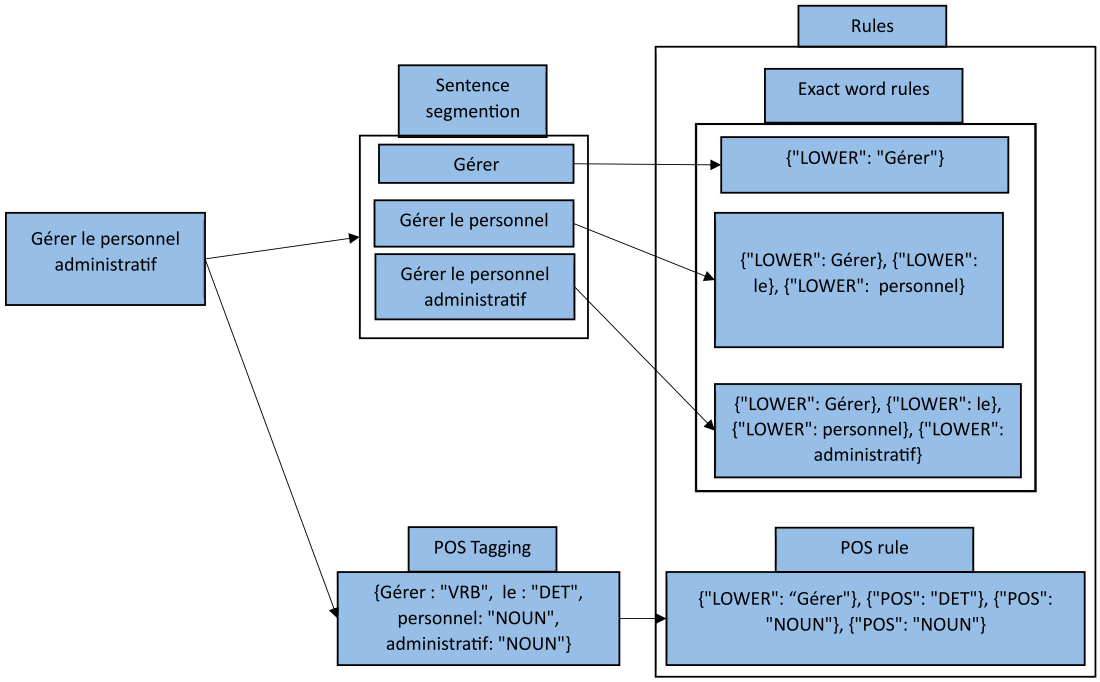


Fig. 3. Example of rule extraction from a hard skill using exact word-based rule building and POS-tagging based rule building A rule pattern is of the form: ("LOWER": WORD1, "LOWER": WORD2). This notation is designed for the Spacy rule-based NER. Given this rule sub sentences that are of the form "WORD1 WORD2" will be extracted.

were combined with relation extraction to create a dataset that includes hard skills, soft skills, economic sector, recruiter company, function, and courses for each job title.

The weight assignment process is based on the frequency by which a triple occurs in its corresponding source data (i.e. job offers or courses). The weight of a triple occurring is the normalized count of the triple normalized by the number of occurrences where the head and relation co-occur. This normalization process is used to assign a weight value in [0, 1]. Mathematically, each triple (h,r,t) can be calculated by the equation $p_{(h,r,t)} = N_{(h,r,t)}/N_{h,\tau(t)}$, where $N_{(h,r,t)}$ represents the number of job offers or courses that contain (h, r, t) triples, and $N_{h,\tau(t)}$ represents the number of job offers or courses where entity h and tail type $\tau(t)$ co-occur. When (h, r, t) is a valid triplet, the condition $p_{(h,r,t)}$ belongs to the set $(0, 1]$ holds; otherwise, $p_{(h,r,t)} = 0$.

3.4. Knowledge graph quality evaluation

The quality of a knowledge graph (KG) is a crucial factor in determining its suitability for application usage. Assessing data quality poses a challenge for both data consumers and providers (Wang and Strong, 1996). Various dimensions of quality, such as completeness, accuracy, provenance, and accessibility, have been identified in the literature as criteria for evaluating KG quality (Issa et al., 2021). However, these dimensions can be subjective, as completeness depends on the specific needs of the data consumer, which can vary significantly. In this study, our focus is on evaluating the completeness of our KG to determine its suitability for applications. Completeness, as a data quality metric, quantifies the extent of information present in a specific dataset (Zaveri et al., 2016). In real-world scenarios, incomplete data can result in the loss of important information and lead to inaccurate analysis (Zaveri

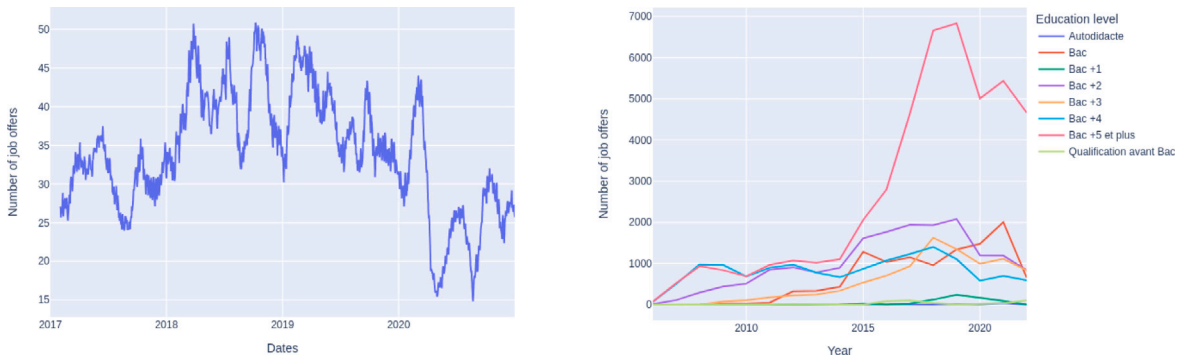


Fig. 4. Numbers of job offers per year from 2017 up until 2020 (left). Numbers of job offers versus the required education level per year starting 2005 up until 2022 (right). The education levels are: 5 years or more of higher education (Bac+ 5 et plus), 4 years of higher education(Bac+ 4), 3 years of higher education (Bac+ 3), 2 years of higher education (Bac+ 2), 1 year of higher education (Bac+ 1), high school diploma (Bac), qualifications before finishing high school (Qualification avant bac) and self-taught (Autodidacte).

et al., 2013). Furthermore, in terms of timeliness, incompleteness hampers the ability to have all the necessary information at the appropriate moment (Issa et al., 2021).

We calculate the metric for completeness using the following formula:

$$\frac{\text{numbers of properties of an instance}}{\text{total number of mandatory properties}} \tag{1}$$

Mathematically, let $\mathcal{G} = (\mathcal{V}, \mathcal{R}, \mathcal{E})$ be a KG defined as a triple, where $\mathcal{V}$ is the set of entities, $\mathcal{R}$ is the set of relations and $\mathcal{E}$ is the set of triples. Let $\mathcal{G}_\mathcal{T} = (\mathcal{V}_\mathcal{T}, \mathcal{R}_\mathcal{T}, \mathcal{E}_\mathcal{T})$ be its corresponding meta-KG that connects the different entity types $\mathcal{V}_\mathcal{T}$ using relations $\mathcal{R}_\mathcal{T}$ to create meta-triples $\mathcal{E}_\mathcal{T}$. Our quality metric can be defined as follows for the set of all entities:

$$H = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \frac{\sum_{((r',u') \in \mathcal{N}_\mathcal{T}(\gamma(v)))} \mathbb{1}_{\exists (r,u) \in \mathcal{N}(v), \gamma(r) = r' \wedge \gamma(u) = u'}}{|\mathcal{N}_\mathcal{T}(\gamma(v))|} \tag{2}$$

where $\gamma : \mathcal{V} \rightarrow \mathcal{V}_\mathcal{T}$ is a mapping that gives the type of the input entity, $\mathcal{N}(x)$ are the neighbors of entity x in $\mathcal{G}$, $\mathcal{N}_\mathcal{T}(x')$ are the neighbors of entity type x' in $\mathcal{G}_\mathcal{T}$, $\mathbb{1}_{\text{condition}}$ is the characteristic function which equals 1 if the condition is true and 0 otherwise.

3.5. Results

3.5.1. Descriptive statistics

We conduct a descriptive analysis in order to explore our data and get a feel of how the Moroccan job market evolves and how education plays into that. In Fig. 4, we observe the number of job offers per year from 2017 up until 2020. We see that the number of job offers dropped significantly on March 2020. On the left, the plot showcases the numbers of job offers with their required education level per year starting 2005 up until 2022. We see a significant rise in demand for 5 years or more of higher education compared to the others. We also spot how high school diplomas and qualifications before finishing high school are diminishing from the job market demand. More plots of our descriptive analysis can be found on our online platform (http://jobed.datanets.org/).

3.5.2. Knowledge graph summary description and quality evaluation

The completed knowledge graph (KG) obtained from the process illustrated in Fig. 1 and discussed in the preceding section comprises a total of 55718 nodes. Among these nodes, 9736 represent job titles, 21 represent soft skills, 32824 represent hard skills, 35 represent locations, 2714 represent recruiters, 133 represent sectors, 75 represent functions, and 10180 represent MOOCs. These nodes are connected through 1296374 relations of 12 types. The location entities represent the cities where a company that posted the job offer is located. The recruiters' entities represent the names of the companies that posted the job offers. The sectors nodes represent the economic sectors that are presented on the website Rekrute. The connections established between job titles and instances of entities belonging to the mentioned types are semantically meaningful. These relationships are categorized under the relationship type 'belongs_to,' which links the job titles as head entities to the companies as tail entities. This relationship expresses the fact that a job belongs to a specific company. The relation 'requires' links the jobs as head entities to the skills as tail entities express the skills required for a certain job. The relation 'located' links the jobs as head entities to the locations as tail entities and "has" links the job titles as head entities to their functions. Other types of relations link the skills to each other (such as 'co_occurs_with' that links the skill entities with each other based on their co-occurrence in the same job offer) or to other types of instances (such as "acquired_by" that links the skills as the head entity and the course as the tail entity). The relation "provides" links the MOOCs as head entities with the skills as tail entities to express the skills provided by a certain course and "favors" links the MOOCs as head entities with the job titles as tail entities. The core metamodel of our knowledge graph JobEdKG is depicted in Fig. 5.

The quality of the KG was evaluated with the formula aforementioned and resulted in the schema of our KG being 75% complete.

3.5.3. Insight extraction

Obtaining high-level insights is a crucial undertaking for monitoring and comprehending the labor market. We extract the most relevant soft skills, hard skills and courses based on the confidence scores assigned as shown in Fig. 6. We choose one of the most occurring jobs in our knowledge graph "Analyste Programmeur" (Analyst Programmer) and we extract the hard skills, soft skills and courses with their confidence scores from JobEdKG. We see that the most relevant hard skills for this job are databases (base de données), C# programming language and MySQL. For the soft skills, novelty research (recherche de nouveauté) and the will to persuade (volonté de persuasion) have the best confidence scores in association with the chosen job title. We also observe that the most relevant courses for this job are management courses such as "Cloud SQL with Terraform" and other programming courses such as "Clean your data with Python"(Nettoyer vos données avec Python).

3.5.4. KG visualization and navigation

In order to visualize and navigate JobEdKG, we built an online platform that allows for visualizing the existing links and the inferred ones based on the model in addition to multiple other features. Furthermore, it offers a user-friendly interface that allows easy exploration and navigation of JobEdKG. Users can explore the graph by clicking on nodes and following the edges to related concepts. The platform also provides search functionality, allowing users to quickly find specific nodes or edges and displays the confidence score for each searched triple. Additionally, users can filter the graph based on specific criteria,

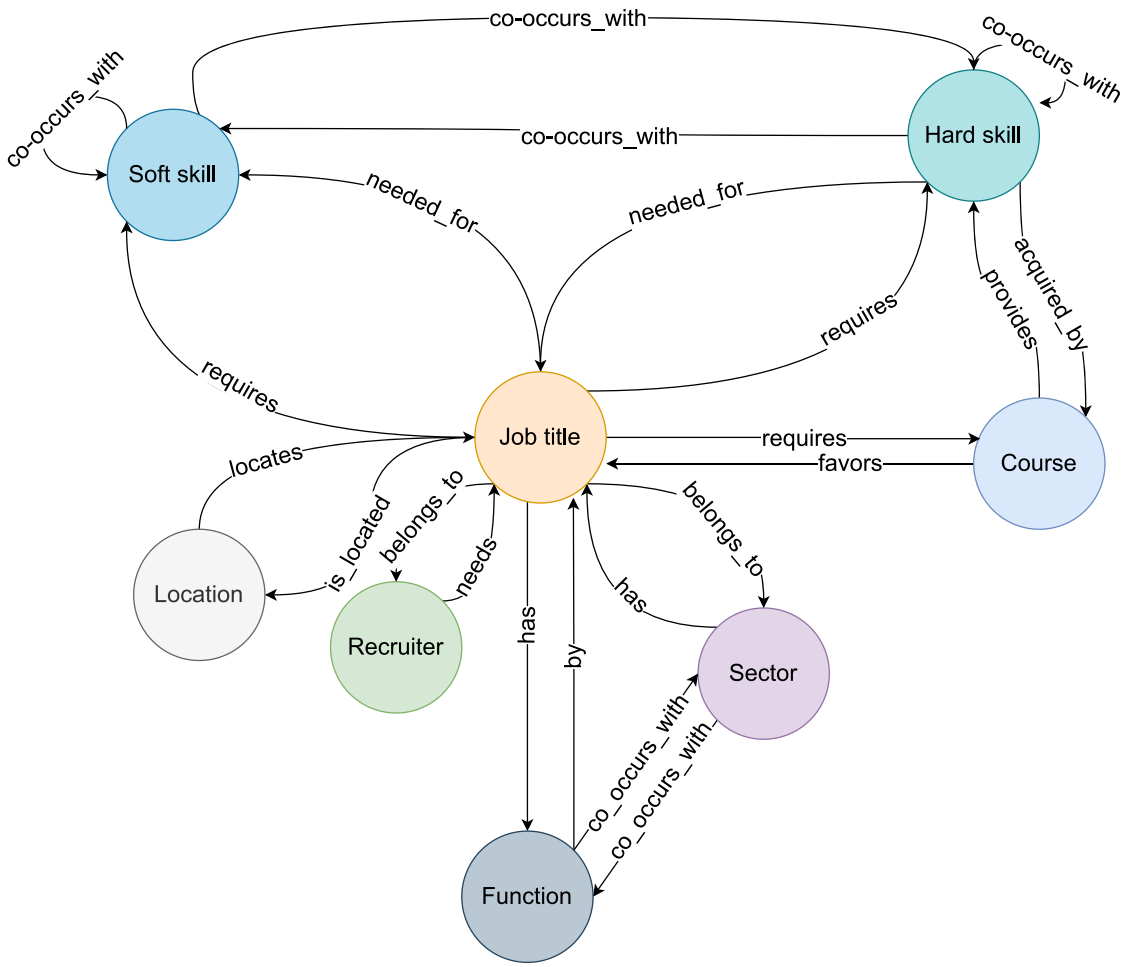


Fig. 5. Metamodel of JobEdKG.

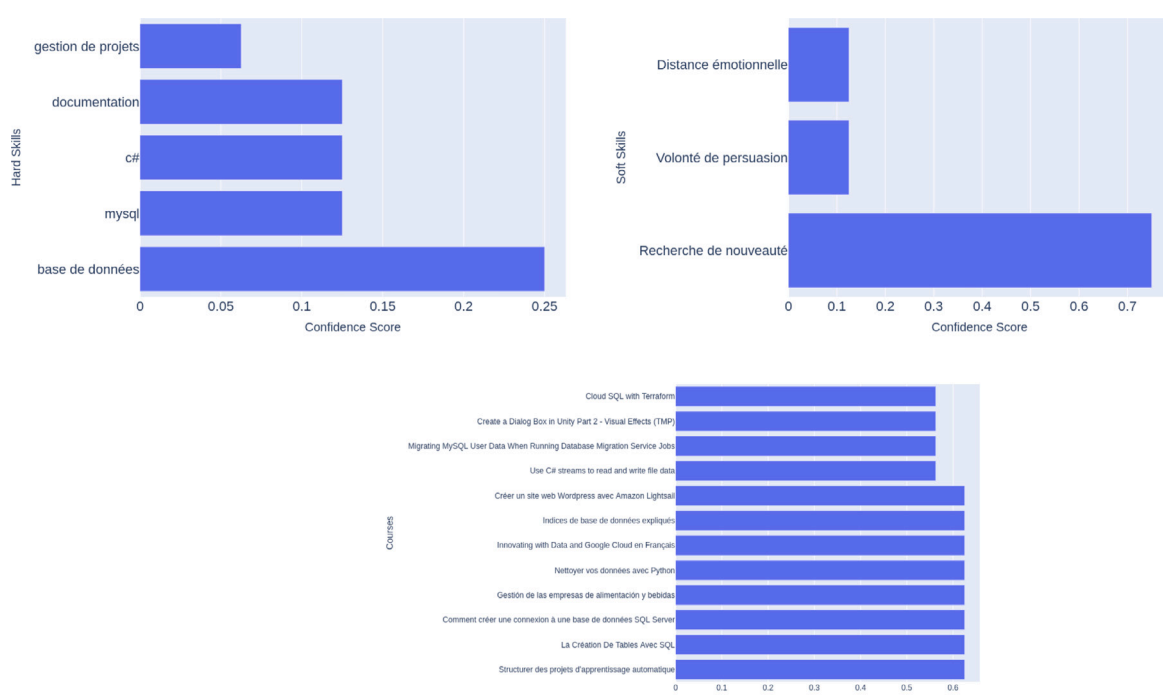


Fig. 6. Top hard skills, soft skills and courses and their confidence scores for the job "Analyst Programmer".

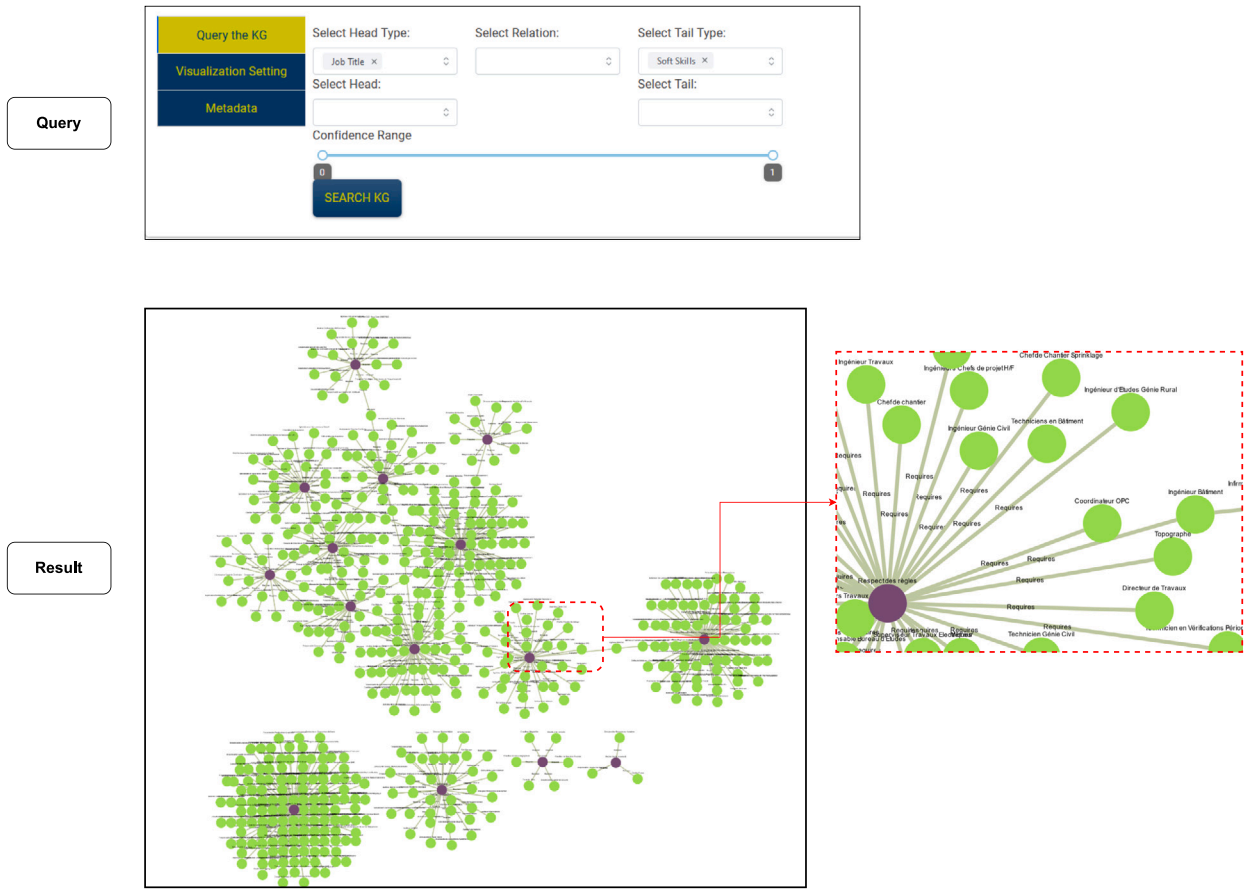


Fig. 7. KG visualization using our online platform.

such as node type or relationship type, to focus their exploration. The platform can be found at http://jobed.datanets.org/. A visualization is showcased in Fig. 7. In the illustration, we showcase the selected head entity types (job title) and the selected tail entity type (soft skills). This query will enable us to visualize all the triples of the KG that have the selected head and tail types. We use the "Cose" display layout in order to showcase the clustering of the nodes. The platform also allows its users to navigate through the model the links predicted by the confidence prediction model. This is discussed in further detail in the next sections.

4. Knowledge graph completion

The constantly changing nature of the labor market poses a challenge when modeling skills and occupations. In this section, we investigate the practical application of our data-driven constructed KG. Our focus is on uncovering new connections between different entities by utilizing the structure of our knowledge graph (KG). To achieve this, we use link confidence prediction models that quantify the association between skill and occupation nodes. Confidence score prediction allows us to discover and quantify the strength of previously unknown connections that were not initially present in our KG.

After extracting the triple and assigning confidence scores to each triple. The resulting KG $\mathcal{G}$ can be seen as an uncertain KG (UKG) (Chen et al., 2019) since each triple $l = (h, r, t) \in \mathcal{G}$ has an associated confidence score $s_l \in [0, 1]$ which quantifies the fact's occurrence likelihood. There are multiple approaches that are used to model UKGs.

In this work, we opted for two baselines to predict confidence. BEURrE (Chen et al., 2021) and UKGE (Chen et al., 2019) in order to compare their performance. The former is an advanced approach, which uses a probabilistic interpretation of geometric concepts, such as volume and intersection in order to predict the confidence scores. The latter adapts an existing score function used for deterministic KGs and maps its value to $[0, 1]$ in order to transform its prediction into a confidence score.

4.1. Embedding layer

The embedding layer is used to represent entities and relations in a lower-dimension space. Properties of this space will be exploited in later stages to calculate the likelihood of the existence of a link between the different entities given the relations. UKGE uses a traditional embedding in $\mathbf{R}^d$. BEURrE uses a geometrically inspired embedding called box embedding, where entities and relations are represented by a hyper-rectangle. In general, a box is an n-dimensional hyperrectangle, i.e. a product of intervals: $[x_i^m, x_i^M]$, where $x_i^m < x_i^M$. This type of box has multiple limitations during training (Dasgupta et al., 2020). Consequently, Dasgupta et al. (2020) proposed a new type of box called the Gumbel box which is defined as follows:

$$\begin{aligned} \mathrm{Box}(X) = \prod_{i=1}^{d} [x_i^m, x_i^M] \text{ where} \\ x_i^m \sim \mathrm{GumbelMax}(\mu_i^m, \beta) \\ x_i^M \sim \mathrm{GumbelMin}(\mu_i^M, \beta) \end{aligned} \tag{3}$$

4.2. Confidence scoring function

UKGE utilized an intuitive approach to predicting confidence scores. In fact, the authors employed the DistMult (Yang et al., 2014) score function $g(l) = \mathbf{r}.(\mathbf{h} \circ \mathbf{t})$ to calculate the plausibility of the triple. In order to obtain confidence scores they mapped the function g to $[0, 1]$ using

Table 1
Distribution of triples in the different KG splits.

Head type	Relation type	Tail type	Test	Train	val
Courses	favors	job titles	57 907	462 890	57 989
Courses	provides	hard skills	0	7396	0
Functions	by	job titles	152	1263	167
Functions	co-occurs with	sectors	13	66	4
Hard skills	acquired by	courses	731	5953	712
Hard skills	co-occurs with	hard skills	112	2165	68
Hard skills	co-occurs with	soft skills	335	3600	282
Hard skills	needed for	job titles	1600	14 547	1569
Job titles	belongs to	recruiters	341	10 629	341
Job titles	belongs to	sector	119	4402	146
Job titles	has	function	1	1581	0
Job titles	is located	locations	101	10 556	95
Job titles	requires	courses	58 017	462 724	58 045
Job titles	requires	hard skills	1758	14 262	1696
Job titles	requires	soft skills	981	8493	1023
Locations	locates	job titles	1076	8606	1070
Recruiters	needs	job titles	1045	9193	1073
Sector	co-occurs with	function	11	67	5
Sector	has	job titles	450	3735	482
Soft skills	co-occurs with	hard skills	421	3374	422
Soft skills	co-occurs with	soft skills	1	11	3
Soft skills	needed for	job titles	1059	8399	1039
	Total		126 231	104 3912	126 231

Table 2
Summary statistics of the KG.

$\lvert\mathcal{V}\rvert$	$\lvert\mathcal{R}\rvert$	$\lvert\mathcal{E}\rvert$
55 718	12	1 296 374

the sigmoid function: $\sigma(x) = \frac{1}{1+e^{wx+b}}$, where w and b are trainable parameters. Consequently, the confidence score for a triple l is predicted as

$$\phi(l) = \sigma(g(l)) \tag{4}$$

$\circ$ is the Hadamard product and $\cdot$ is the dot product.

BEURrE exploited the geometric nature of the embedding and used volume and intersection in a probabilistic framework to predict confidence scores. BEURrE's scoring function is expressed as follows:

$$\phi(l) = \frac{\mathbb{E}[\mathrm{Vol}(f_r(\mathrm{Box}(h)) \cap g_r(\mathrm{Box}(t)))]}{\mathbb{E}[\mathrm{Vol}(g_r(\mathrm{Box}(t)))]} \tag{5}$$

where f_r and g_r are relation-specific transformations performed on the boxes of the heads and tails respectively. Each box $\mathrm{Box}(X)$ is a Gumbel box that is parameterized using the center $\mathrm{cen}(\mathrm{Box}(X)) \in \mathbb{R}^d$ and offset $\mathrm{off}(\mathrm{Box}(X)) \in \mathbb{R}^{d+}$, with the location parameters being given by

$$\begin{aligned} \mu_i^m &= \mathrm{cen}(\mathrm{Box}(X)) - \mathrm{off}(\mathrm{Box}(X)) \\ \mu_i^M &= \mathrm{cen}(\mathrm{Box}(X)) + \mathrm{off}(\mathrm{Box}(X)) \end{aligned} \tag{6}$$

The authors of Chen et al. (2021) performed transformations f on the Gumbel boxes which are parameterized by a translation vector $\tau \in \mathbb{R}^d$ and a scaling vector $\Delta \in \mathbb{R}^{d+}$ such that

$$\begin{aligned} f(\mathrm{cen}(\mathrm{Box}(X)); \tau, \Delta) &= \mathrm{cen}(\mathrm{Box}(X)) + \tau \\ f(\mathrm{off}(\mathrm{Box}(X)); \tau, \Delta) &= \mathrm{off}(\mathrm{Box}(X)) \circ \Delta \end{aligned} \tag{7}$$

where $\circ$ is the Hadamard product. Separate transformations are used for the head and tail entities of a relation, which we denote f_r and g_r, and omit the explicit dependence on the learned parameters τ and Δ.

4.3. Training objective

The previously mentioned parameters are trained using a supervised objective which minimizes the mean squared error between the predicted score and the ground truth score. We follow the same training process as Chen et al. (2021), where scores are calculated using the scoring function for a set of positive (observed) triples $\mathcal{L}^+$ and a set of generated negative triples $\mathcal{L}^-$. These negatives are generated by either corrupting the head or the tail. For each fact $l \in \mathcal{L}^+$, the goal is to minimize the mean squared error between the ground truth confidence score s_l and the prediction $\phi(l)$, and for each fact $l \in \mathcal{L}^-$ the goal is to minimize the MSE between the ground truth confidence score s_l and 0. The learning objective can be expressed as follows:

$$\mathcal{J} = \sum_{l \in \mathcal{L}^+} |\phi(l) - s_l|^2 + \sum_{l \in \mathcal{L}^-} |\phi(l)|^2 \tag{8}$$

The set $\mathcal{L}^-$ is generated by corrupting the head and the tail for observed facts. Fig. 8 showcases the process of how the BEURrE and UKGE models are trained.

4.4. Inference

After training UKGE and BEUrRE we can easily infer the confidence scores of new facts using their respective scoring functions described in Eq. (4) and Eq. (5). This is done by embedding the different entities and relation of a triple in the appropriate embedding space for each scoring function and then applying the scoring function to the embeddings. Fig. 8 shows the inference process for a triple.

5. Experiments

5.1. Dataset

Our knowledge graph is composed out of multiple entities and relations pertaining to the job market. Table 2 shows the summary statistics of the KG. Table 1 gives the number of each meta-triple in the different splits.

5.2. Implementation details

We trained our model on a GEFORCE RTX 3070. We utilize the Adam optimizer and fine-tune the hyperparameters using grid search based on the performance on the validation set. The hyperparameter search range and the best hyperparameters are detailed in Tables 3 and 4, respectively. Early stopping is employed based on the same metric, with patience of 50 epochs. The extracted triples are divided into three sets: training, validation, and testing. The distribution of these sets is as follows: 85% for training, 7% for validation, and 8% for testing. Each experiment is repeated five times, and the average results are reported.

Table 3
Search Space for hyper-parameters.

Hyper-parameters	BEURrE	UKGE
Learning rate lr	$\{0.001, 0.0001, 0.00001\}$	$\{0.001, 0.0001, 0.00001\}$
Embedding dimension d	$\{30, 64, 128, 300\}$	$\{50, 100, 300\}$
Batch size b	$\{256, 512, 1024, 2048, 4096\}$	$\{256, 512, 1024, 2048, 4096\}$
Gumbel box temperature β	$\{0.1, 0.01, 0.001, 0.0001\}$	–
L2 regularization λ	$\{0.001, 0.01, 0.1, 1\}$	–

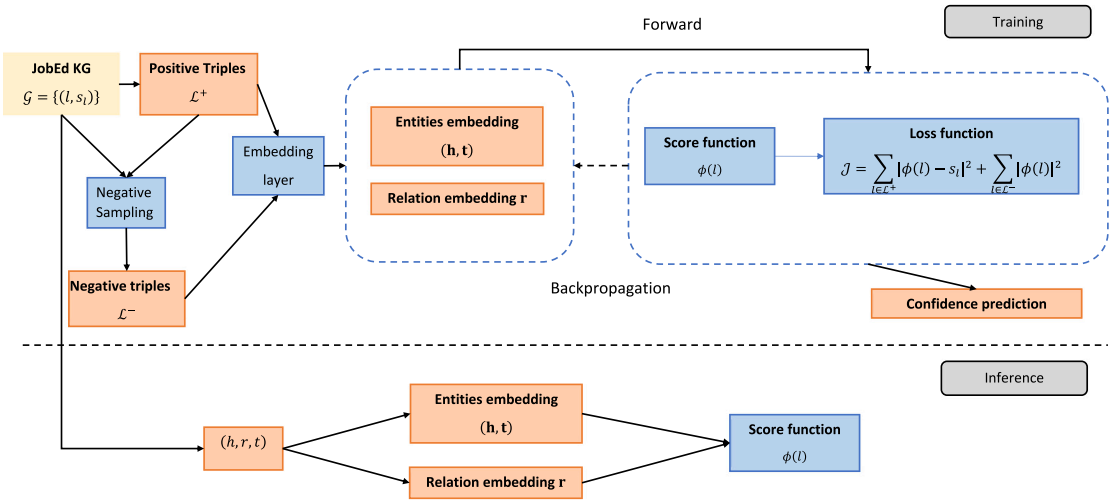


Fig. 8. Training and inference process for BEURrE and UKGE models. The orange components represent the results and the blue components represent the processes. The inference process is described for a particular triple.

Table 4
Optimal hyper-parameters for different tasks.

Task	lr	b	d	β	λ
BEURrE	0.0001	4096	64	0.01	0.001
UKGE	0.0001	4096	100	–	–

Table 5
Results of fact confidence prediction ($\times 10^{-2}$).

Model	MSE	MAE
BEURrE	0.07	0.64
UKGE	0.10	0.92

5.3. Baselines

We compare two baselines BEURrE and UKGE for the task of confidence prediction on JobEdKG.

- BEURrE (Chen et al., 2021): This model uses box representations of entities and scalar representations of relations. It transforms the embedding of entities using affine transformations using relation vectors.
- UKGE (Chen et al., 2019): This model uses a deterministic scoring function that maps to $[0, 1]$ in order to predict the confidence score of a triple.

5.4. Experimental results

Table 5 presents the outcomes for the confidence prediction task on the JobEdKG, demonstrating the notable results achieved. These outcomes can be attributed to the completeness of the KG, as the extracted triples encompass a wide array of potential connections between entities, incorporating diverse levels of confidence, and reducing the number of false negatives that can be caused by negative sampling. We observe that the results of BEURrE are slightly better than UKGE. This is expected since the former achieves state-of-the-art performance in multiple datasets.

5.5. Ablation study

Table 6 gives results for each meta-triple in the test dataset using UKGE and BEURrE models. We notice that BEURrE consistently outperforms UKGE on the different meta-triples. In addition, meta triples with lesser frequency (see Table 1) have a lower performance. This is analogous to the effect of imbalance in supervised classification tasks, which can result in forcing the model to ignore minority classes.

Fig. 9 shows the distribution of entity embeddings with and without relation transformation. We can see that the relation transformations (described in Eq. (7)) improve the discriminative power of entity embeddings, which justifies the use of the added computational complexity of the transformation.

5.6. Case study

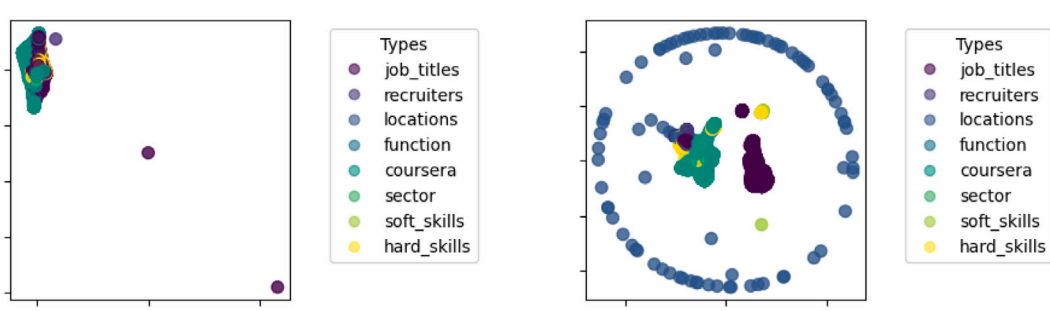
The constantly changing nature of the labor market poses a challenge when modeling skills and occupations. In this section, we analyze the outcomes of our trained model on JobEdKG. Our primary objective is to uncover fresh connections between skills, courses, and occupations by utilizing the structure of our KG. Table 7 showcases the results of the predicted entities based on the BEURrE model for the jobs 'Management Control' where we see top hard skills such as 'accounting procedures' and soft skills such as 'need for objectivity' as well as data analysis related courses such as 'Analyze City Data Using R and Tableau'. The occupation 'Executive Assistant' has novel inferred entities as well such as 'perform an administrative task' as a hard skill and soft skills such as 'humility' as well as novel associated courses such as 'Business English: Final Project'.

Table 6
Experimental results of UKGE and BEURrE on the test dataset.

Head type	Relation type	Tail type	UKGE		BEURrE	
			MAE	MSE	MAE	MSE
Coursera	favors	job titles	0.0060	0.0002	0.0057	0.0002
Function	by	job titles	0.0449	0.0070	0.0270	0.0070
Function	co-occurs with	sector	0.0411	0.0032	0.0371	0.0028
Hard skills	acquired by	courses	0.0400	0.0080	0.0271	0.0045
Hard skills	co-occurs with	hard skills	0.6326	0.4115	0.0335	0.0052
Hard skills	co-occurs with	soft skills	0.0250	0.0033	0.0186	0.0019
Hard skills	needed for	job titles	0.0377	0.0083	0.0298	0.0062
Job titles	belongs to	recruiters	0.0693	0.0119	0.0460	0.0065
Job titles	belongs to	sector	0.0359	0.0089	0.0224	0.0016
Job titles	has	function	0.0879	0.0077	0.0847	0.0072
Job titles	is located	locations	0.0170	0.0008	0.0169	0.0008
Job titles	requires	courses	0.0062	0.0003	0.0060	0.0003
Job titles	requires	hard skills	0.0601	0.0192	0.0600	0.0192
Job titles	requires	soft skills	0.0066	0.0006	0.0066	0.0006
Locations	locates	job titles	0.0078	0.0015	0.0078	0.0015
Recruiters	needs	job titles	0.1094	0.0313	0.0720	0.0165
Sector	co-occurs with	function	0.0428	0.0031	0.0321	0.0021
Sector	has	job titles	0.0325	0.0062	0.0243	0.0047
Soft skills	co-occurs with	hard skills	0.0197	0.0036	0.0150	0.0021
Soft skills	co-occurs with	soft skills	0.8096	0.6555	0.1733	0.0300
Soft skills	needed for	job titles	0.0058	0.0005	0.0057	0.0003

Table 7
Existing and inferred soft skills, hard skills and courses for the jobs 'Management Control' and 'Executive Assistant'.

Job title		Hard skills	Soft skills	Courses
Management control	**Existing entities**	Finance; Financial analysis; Information Systems; Commerce; Data analysis	Need for objectivity; Rationalism; Will to persuade; Novelty research	Fund an impact project; Analyze data to answer questions; Run the project; Python 101: Develop Your First Python Program; Creating Trust & Support
	Inferred entities	Participate in the definition of the commercial strategy; Accounting procedures; Professional English; Guide the teams	Conventional; Humility, Improvising; Ambition	Speaking to persuade: Motivating audiences with solid arguments and moving language; Advanced Understanding of Stocks and Bond; Analyze City Data Using R and Tableau
Executive assistant	**Existing entities**	provide physical reception; Analyze business indicators; ensure the pre-sale of products; Prepare letters; Ensure a high level of quality	Intuition and spontaneity; Rationalism; Conventional; Will to persuade	Engagement & Nurture Marketing Strategies; Brand Management: Aligning Business, Brand and Behavior; The Roles and Responsibilities of Nonprofit Boards of Directors within the Governance Process
	Inferred entities	Perform administrative tasks; Ensure formal validations, Ensure a quality welcome for customers; Corrective actions; Track achievements	Humility; Ambition, Emotional sensitivity	Advanced Commands in Windows; Business English: Final Project; Leverage a Johari Window to Develop Strong Teams in Miro



(a) Distribution of entity embeddings without relation transformation.

(b) Distribution of entity embeddings with relation transformation.

Fig. 9. t-Distributed Stochastic Neighbor Embedding (t-SNE) visualizations for the distribution of entity embeddings with (Fig. 9(b)) and without relation transformations (Fig. 9(a)).

6. Discussion

The labor market has undergone significant changes in recent years, primarily driven by factors such as increased global competition, a rapidly growing working population, and the widespread adoption of digital technologies. The COVID-19 pandemic has expedited this transformation, as businesses and industries have been forced to swiftly adjust to novel work methodologies and economic circumstances. These

factors have resulted in significant shifts in the types of jobs available and the skills required to succeed in them. Our aim is to explore innovative approaches to matching job seekers with vacancies more effectively by using advanced algorithms and data-driven methods. Specifically, we propose building a knowledge graph that models skills, courses, and occupations in order to better understand the relationships between them. By understanding the relationships between different occupations and the skills required to excel in them, job seekers can better identify opportunities for career growth and development. This paper focuses on constructing a knowledge graph that integrates information from the job market and Massive Open Online Courses (MOOCs). The aim is to provide course recommendations for job seekers, helping them acquire the necessary skills to excel in their desired occupations. Firstly, we extract and process semi-structured data. During this process, we perform skill entity extraction using rule-based methods and we filter the skill entities using a pre-trained sentence acceptability model. For validating the completeness of our KG and its validity for use in applications, we evaluate our KG quality using a KG quality metric in Section 3.5.2. We model our final KG using state-of-the-art KG embedding models. We use the BEURre and UKGE models for uncertain KG embedding to be used for confidence prediction and inferring non-existent links in the KG and their confidence scores. Next, we explore our KG for job-skill and job-course matching as shown in Section 5.

Our methodology can be further complimented by using more elaborate approaches to solve some limitations of the different approaches. Data preprocessing involved several sequential steps, including the extraction of skill entities as outlined in Section 3. We opted for a rule-based method. More refined methods can be employed such as model-based methods that demand explicit annotation from experts or further improvements such as adding more rules and then using pre-trained language models. Furthermore, during the KG construction phase, we opted for the most efficient way to calculate confidence scores described in Section 3 but other methods can be utilized as well.

The methods presented in this context are flexible in terms of data requirements, as long as similar concepts such as occupations and skills, as well as relevant data such as job postings and courses, are accessible. To accommodate the bilingual nature of our framework (English and French) and since the study was done on the Moroccan labor market, we leverage taxonomies like ESCO, ROME, and Jobzilla. ESCO and ROME are established standards and freely available, while other frameworks like O*NET,[10] widely used in the United States, can also be utilized.

To summarize, our research focuses on investigating algorithmic approaches that aim to improve the alignment between individuals and occupations, benefiting both job seekers and recruiters. In order to do this, and differently from previous studies such as de Groot et al. (2021), we propose a general framework to construct a KG that combines both labor market and education entities. Our practical process concludes with careful consideration and also serves the purpose of providing theoretical guidelines.

The primary aim of introducing the framework is not only limited to the construction of a KG based on the Moroccan labor market but also to serve as a point of reference for building a similar knowledge graph in various other countries. Our pipeline could be referred to in other KG construction studies that provide theory support.

We also built an online platform that offers an intuitive and efficient way for users to explore and understand the relationships within JobEdKG. This way job seekers can define the skills and the courses needed for their desired occupations. We also discuss potential future directions for this line of inquiry.

[10] https://www.onetonline.org/

6.1. Future directions

The richness of our KG provides an ideal resource for multiple applications and perspectives:

- Entity resolution: The skill entity extraction process can be improved by adding an entity resolution pipeline that would align similar entities into a single one, which will improve the KG embedding process for better confidence prediction.
- Temporal Uncertain KG: Adding temporality to our KG for better link and confidence prediction could be a promising addition for better accuracy of downstream tasks.
- Recommender system for recruiters and job seekers: Graph-based recommendation systems leverage domain knowledge to generate outcomes. By incorporating semantic relationships within our knowledge graph, we enhance the precision and variety of recommendations for both recruiters and job seekers. Furthermore, our knowledge graph facilitates path-based recommendation, enabling the exploration of underlying patterns and retrieval of pertinent information for generating recommendations.

7. Conclusion

Skill mismatch is a significant issue that is crippling young graduates and by extension world's economy. This research introduces the initial uncertain knowledge graph that combines entities from the education and job market domains. We aimed to unveil the connections between education and the job market as well as predict the possible links between these entities. We propose an automated framework for KG construction from semi-structured data and evaluate this construction using the KG completeness evaluation metric. We approach the task of associating skills with courses and job titles as a link prediction problem. To address this, we employ a cutting-edge uncertain knowledge graph embeddings model that excels in link prediction. This model enables us to infer new connections between entities within the knowledge graph.

CRediT authorship contribution statement

Yousra Fettach: Conceptualization, Software, Formal analysis, Investigation, Methodology, Writing – original draft. **Adil Bahaj:** Conceptualization, Software, Investigation, Methodology, Writing – original draft. **Mounir Ghogho:** Conceptualization, Formal analysis, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the data and the code used in this research.

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